



Introduction to Astrophysics

Cosmology: Beyond the Milky Way

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20th May 2015

What is Cosmology?

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Shape: What happens as you travel across the universe?

Content: What are Dark Matter and Dark Energy?

The Night Sky

The Milky Way





The Night Sky

The Milky Way



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The Milky Way



Ground based telescopes

The Coma Cluster



Ground based telescopes

The Coma Cluster



Space Telescopes

Hubble Extreme Deep Field



Space Telescopes

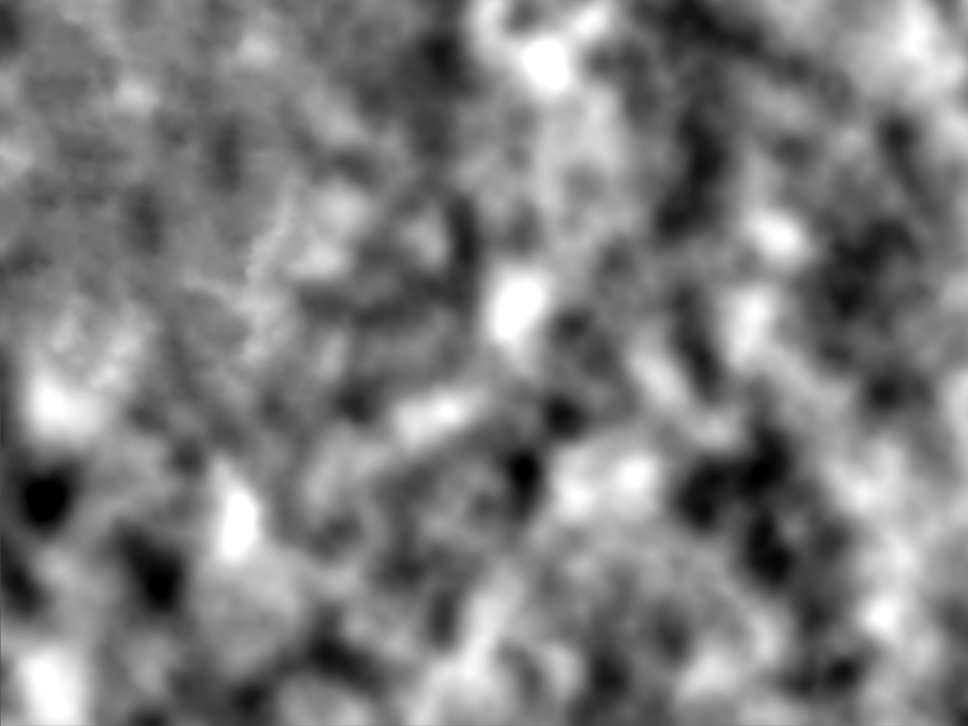
Hubble Extreme Deep Field

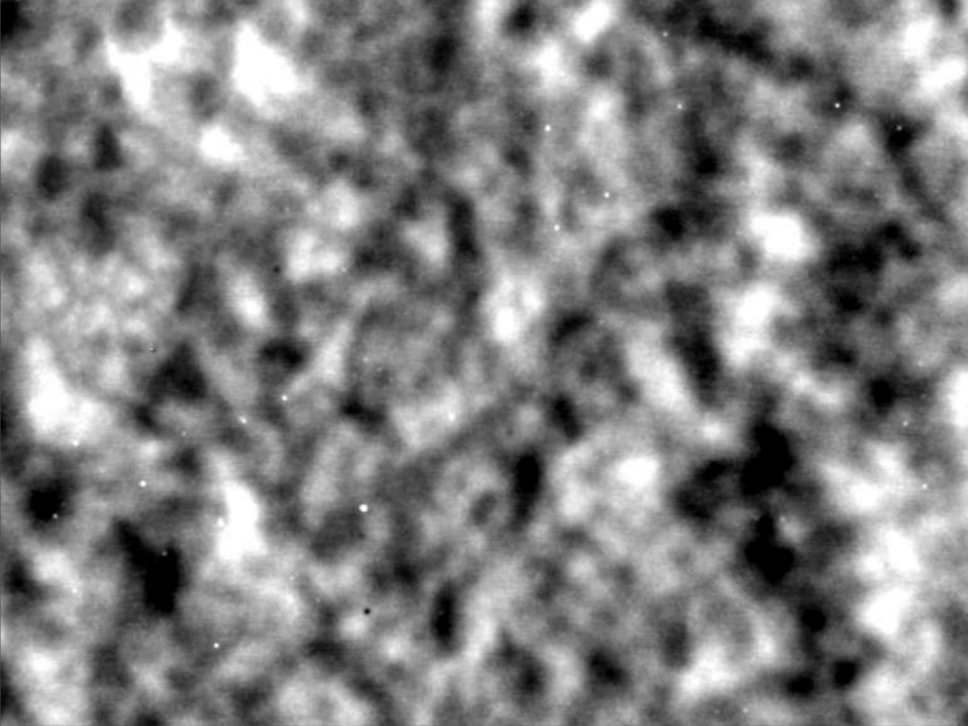


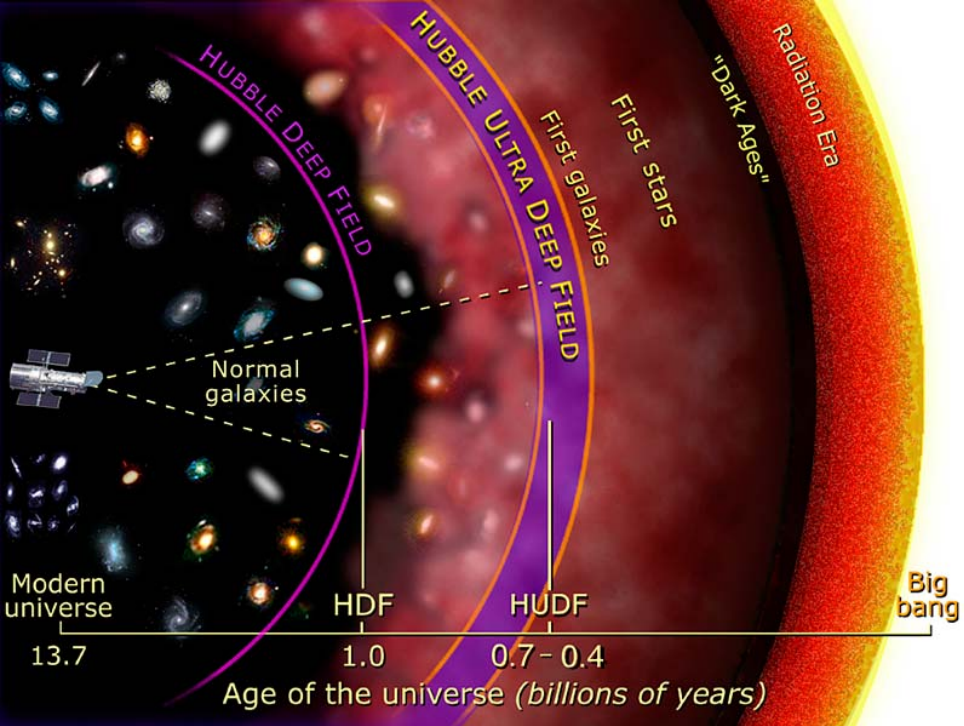
The Dark Ages

The CMB

Cosmic Microwave Background







What is the CMB?

- ▶ In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with “The Big Bang”.

The Big Bang

Galactic Redshift



The Big Bang

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The Big Bang

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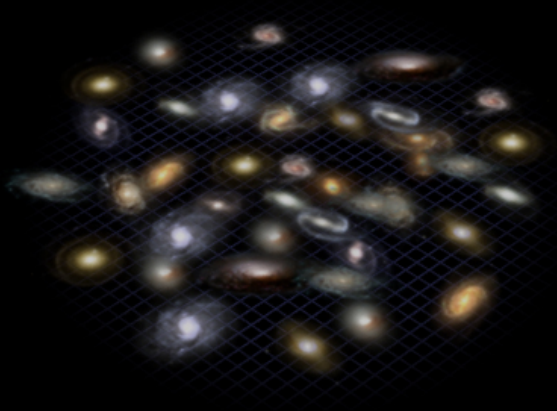
1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away

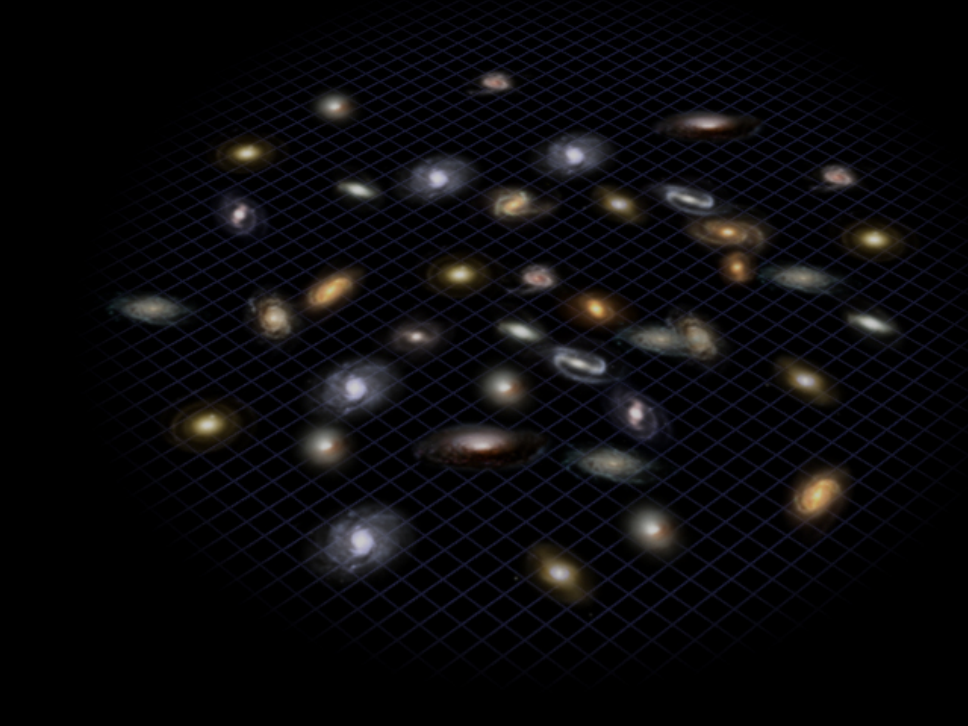
The Big Bang

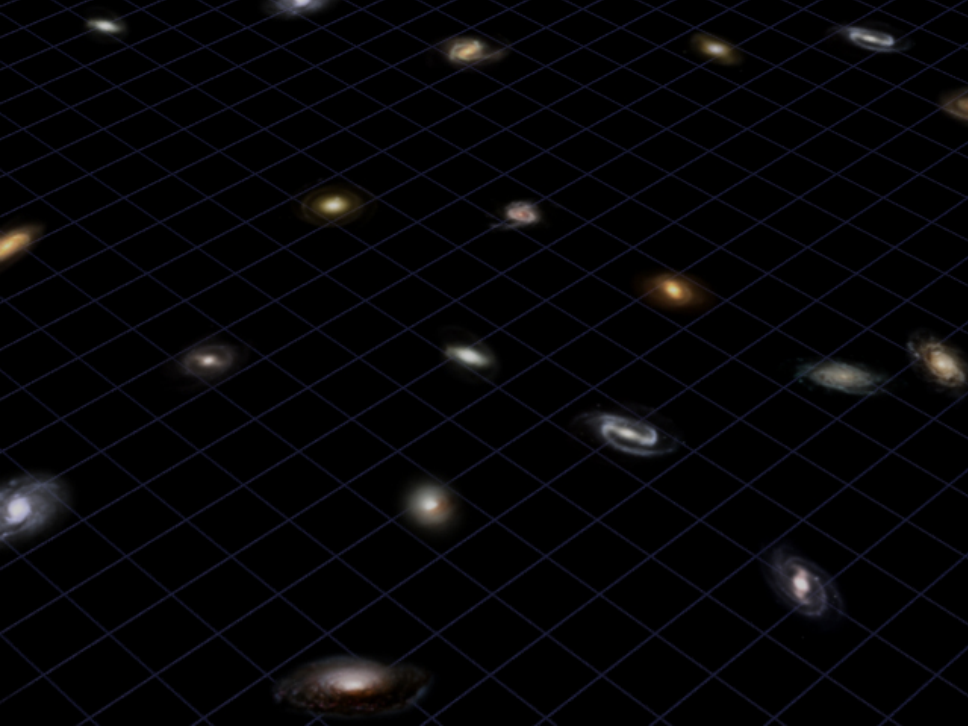
Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

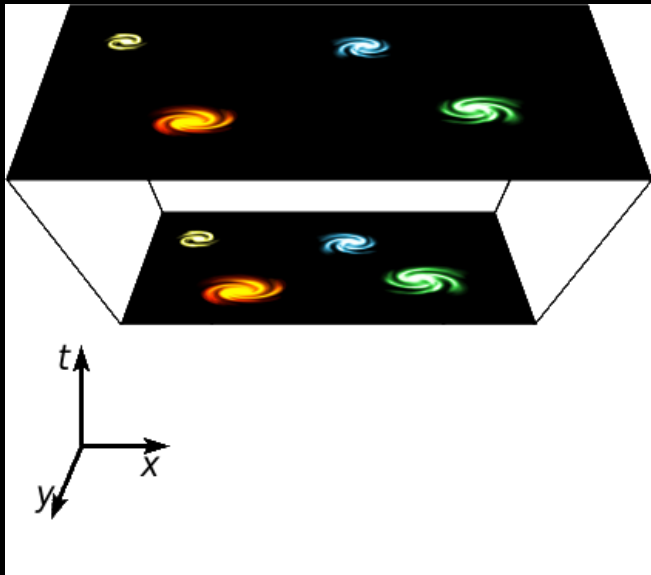






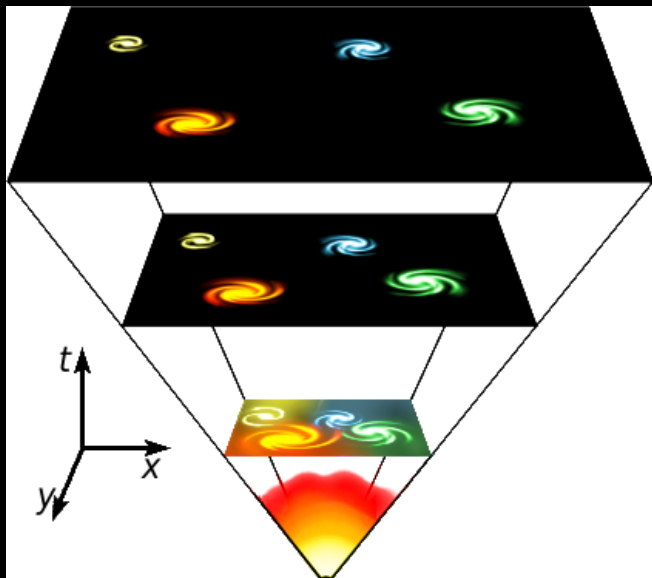
The Big Bang

A stretching universe:



The Big Bang

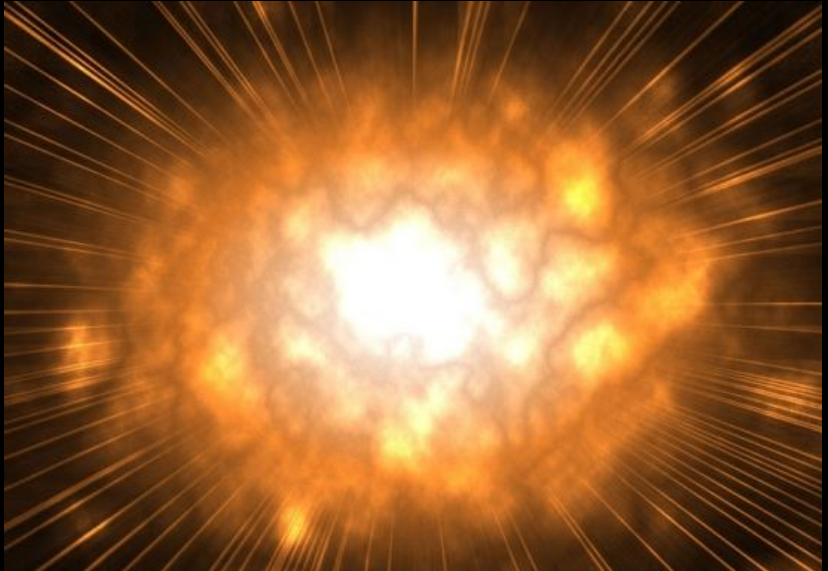
A stretching universe:



The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:



The Big Bang

Common Misconceptions

- ▶ Despite the name, the Big Bang is not an explosion.

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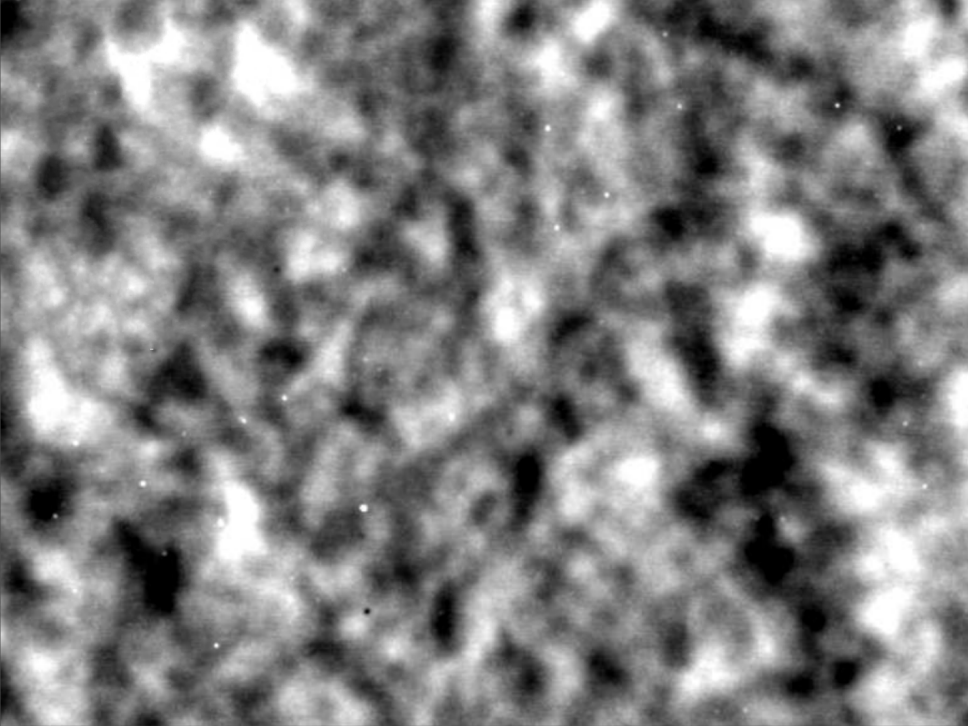
Common Misconceptions

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The Big Bang

Common Misconceptions

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- ▶ The Universe is not necessarily expanding into anything.
- ▶ The big bang didn't begin at a particular point



Current Issues in Cosmology

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Dark Matter

The Problem



Dark Matter

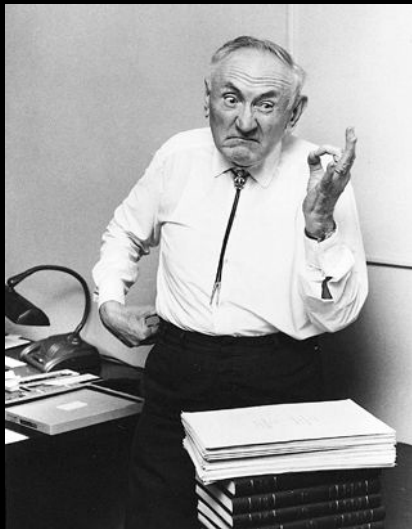
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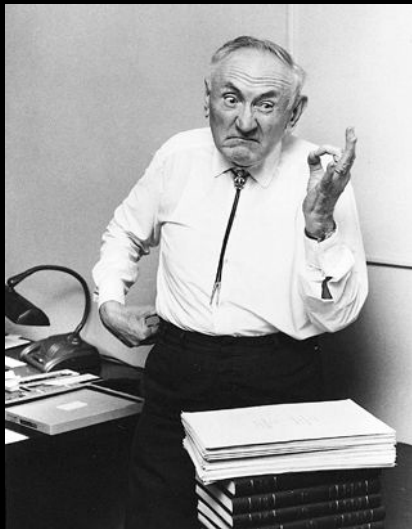
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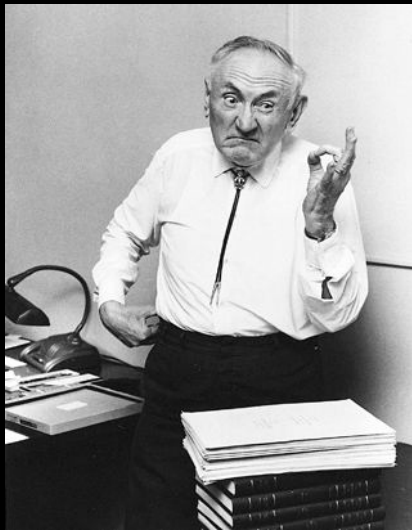
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Dark Matter

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- ▶ Outnumbers traditional matter 7:1



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ESA 1993 MPS for OSIRIS Team MPS/OPOL/MLA/PRD/INT/AS/INAC/SP/IN



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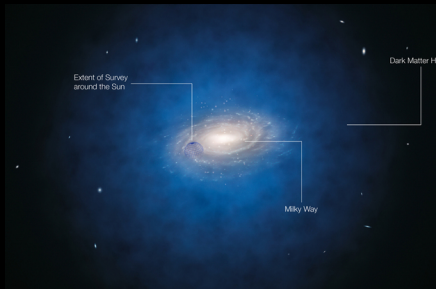
ESA 2013 MPS by OSIRIS-THEMIS/OSIRIS/MASSIMO/ESA/SPACEDOC



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ESA 2013 MPE for COSMOS Team: HP/SPOL/IMM/ARF/ST/AS/SP/SC/SP/CH



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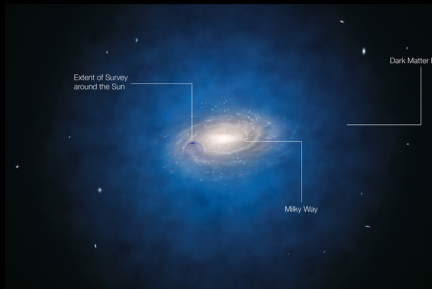
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- ▶ Right now it's really just a name for a phenomenon we don't understand yet.

ESA 2013 MPE by COSIE Team HPF05POLIMARKEDD01ASPACEOPERA



Dark Energy

First things first...

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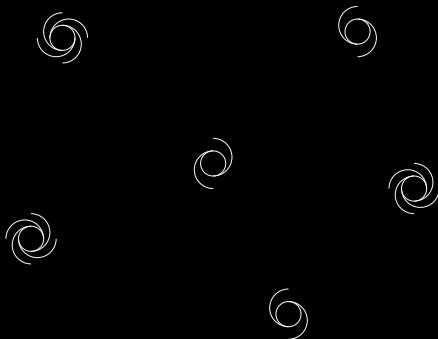
- ▶ There is no *a priori* reason to assume that these are related.
- ▶ Dark matter: 1930's
- ▶ Dark energy: 1990's

Universal consequences of gravity

Thought experiment

Universal consequences of gravity

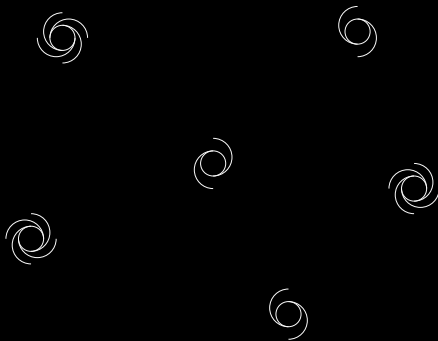
Thought experiment



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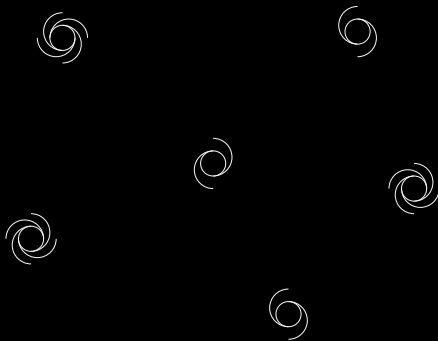
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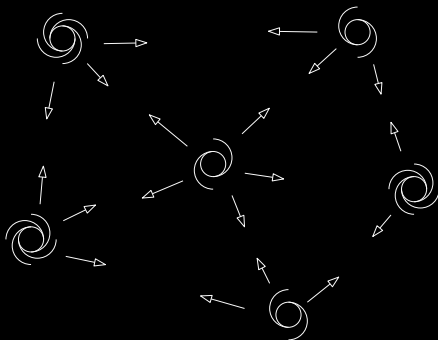
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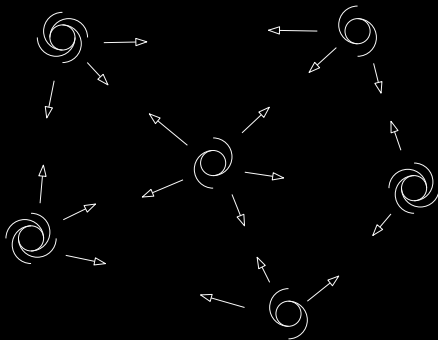
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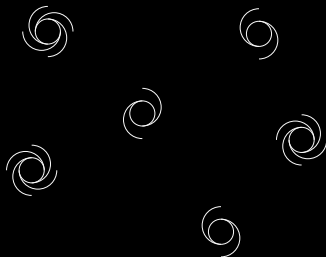
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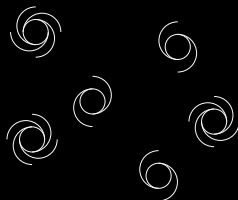
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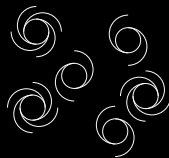
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The consequences for the universe

Reality

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Reality

- ▶ We know that galaxies are currently moving apart.
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- ▶ This suggests two possibilities...

The evolution of the universe

Two possibilities

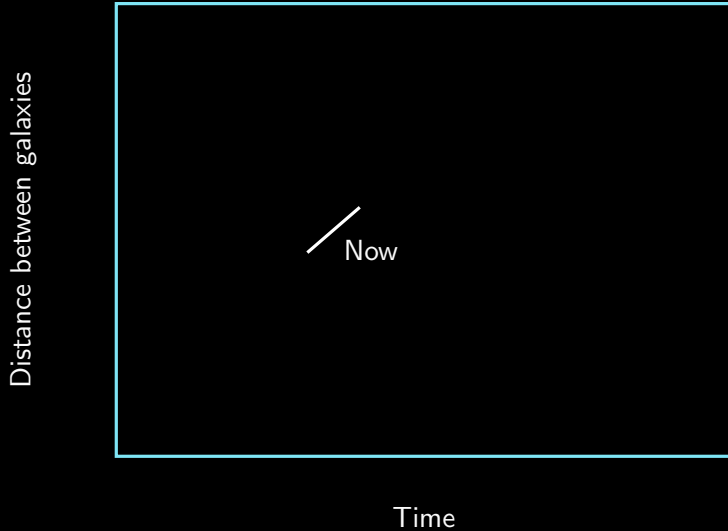
Distance between galaxies



Time

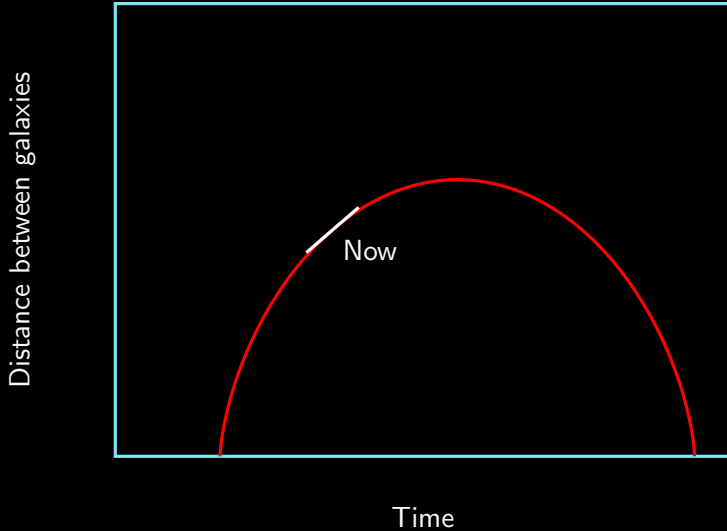
The evolution of the universe

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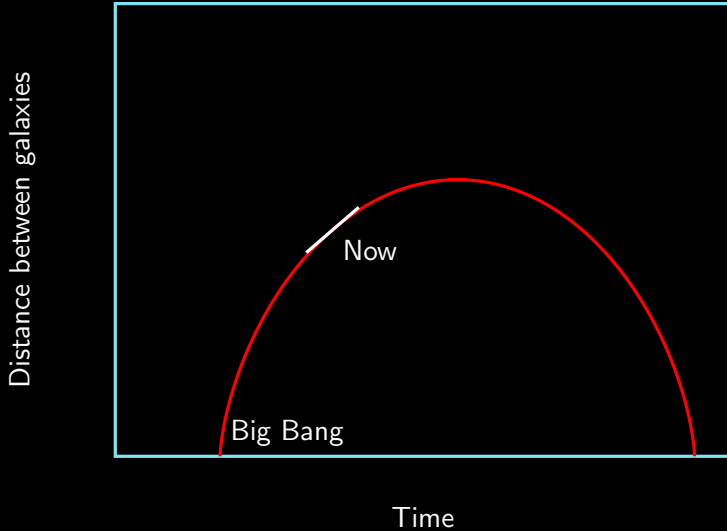
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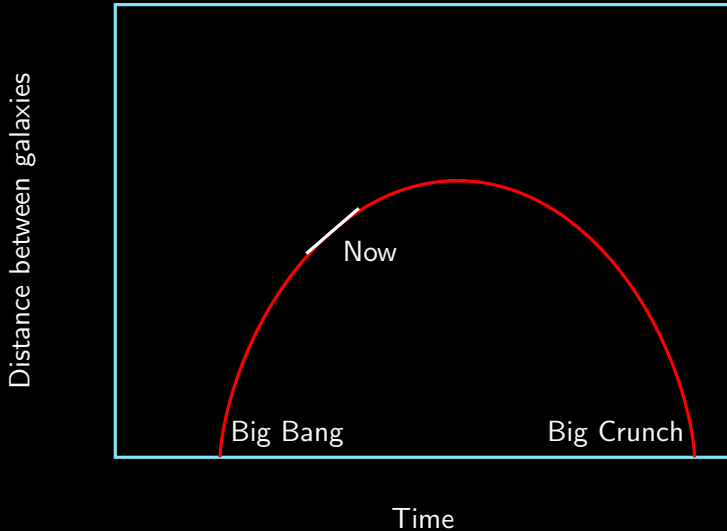
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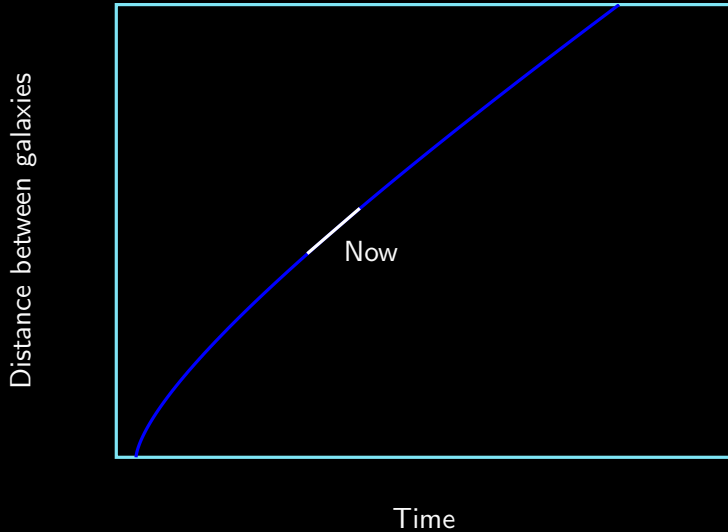
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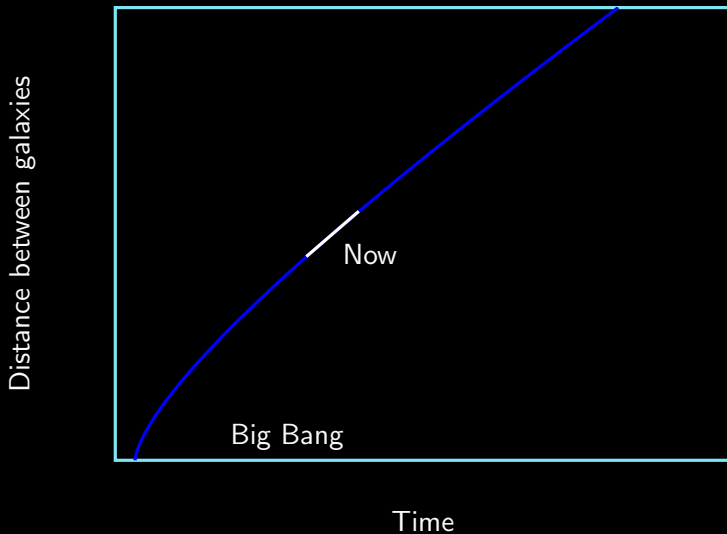
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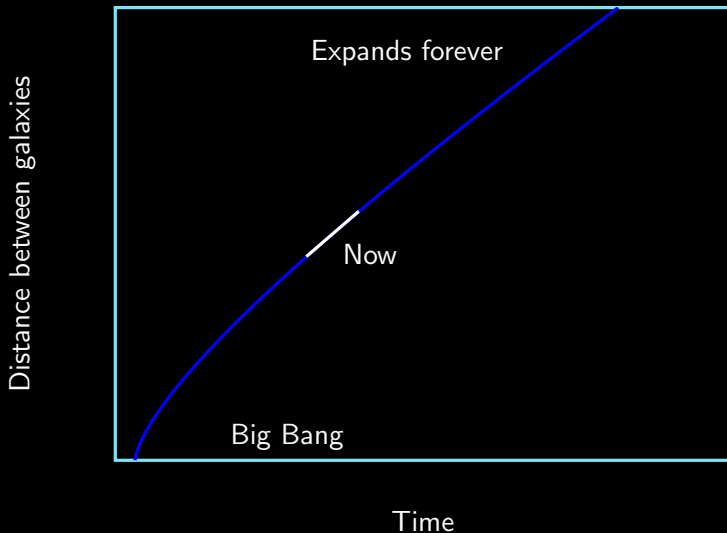
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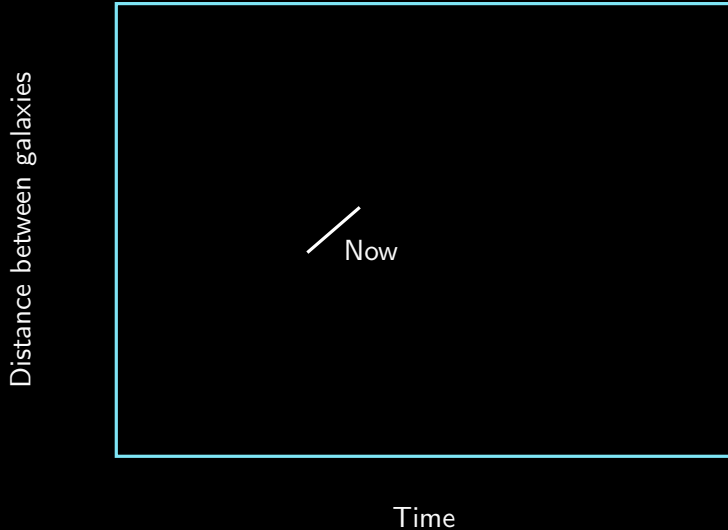
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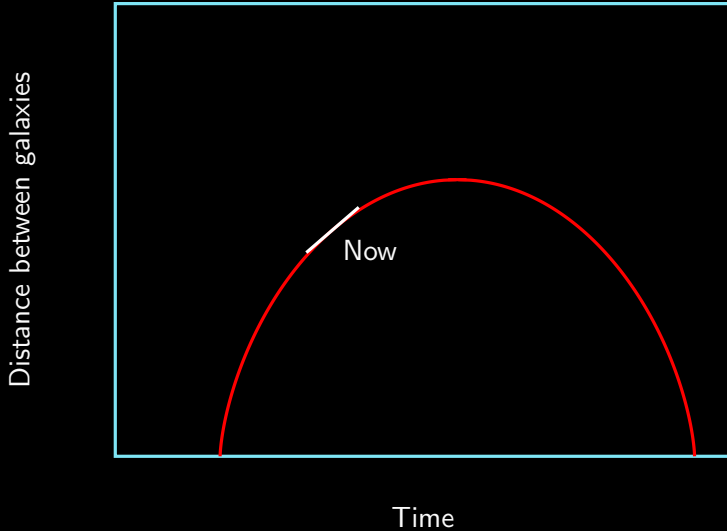
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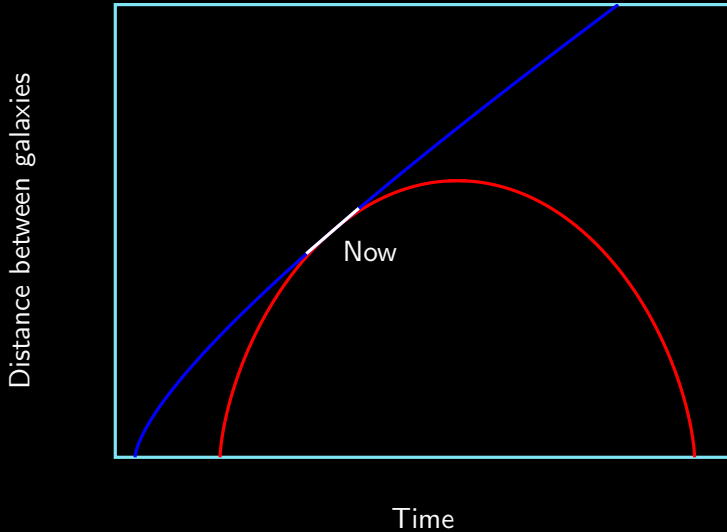
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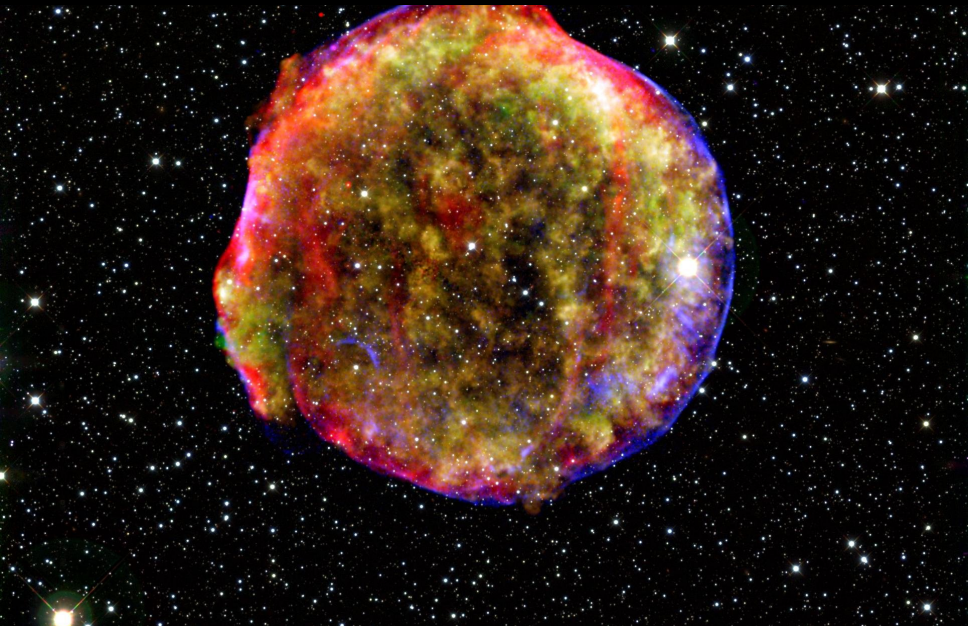
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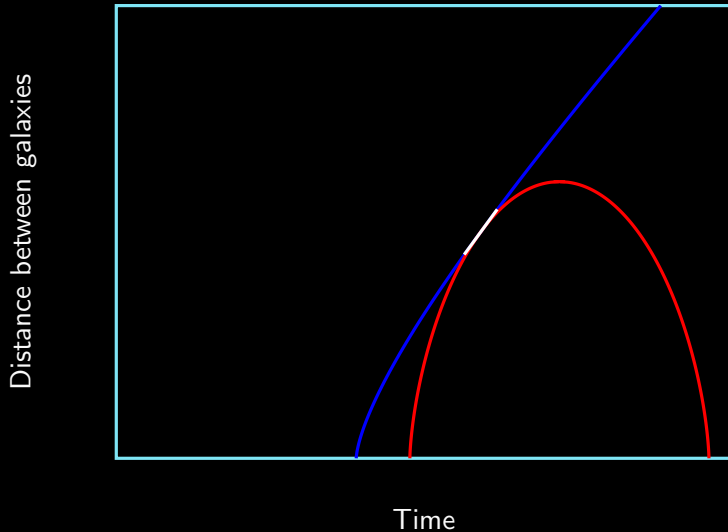
Exploding stars

Type IA Supernovae



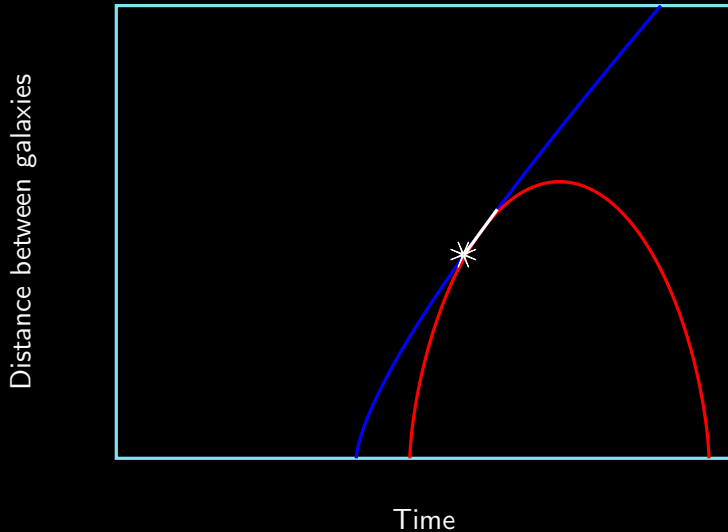
The evolution of the universe

Which is it?



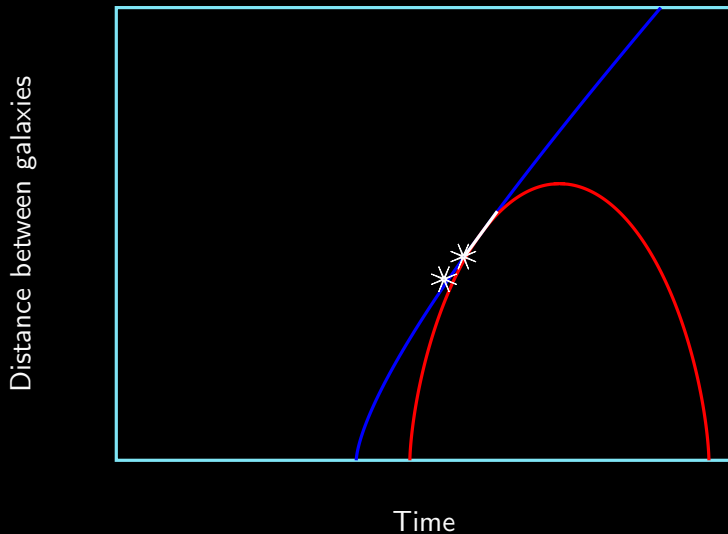
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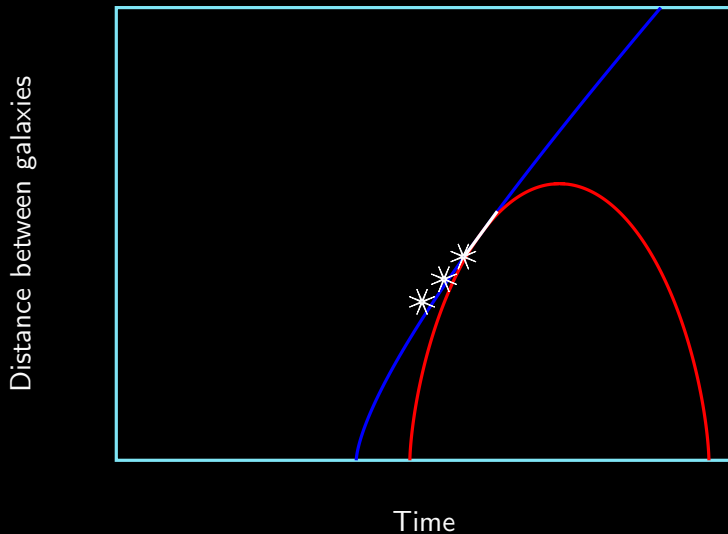
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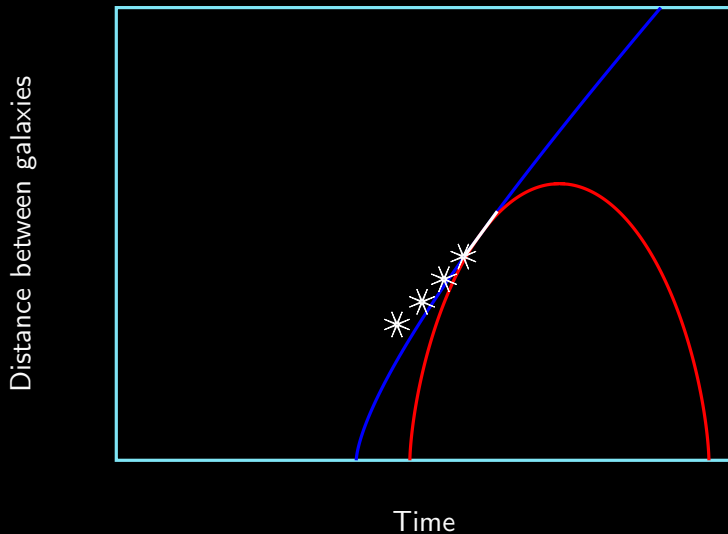
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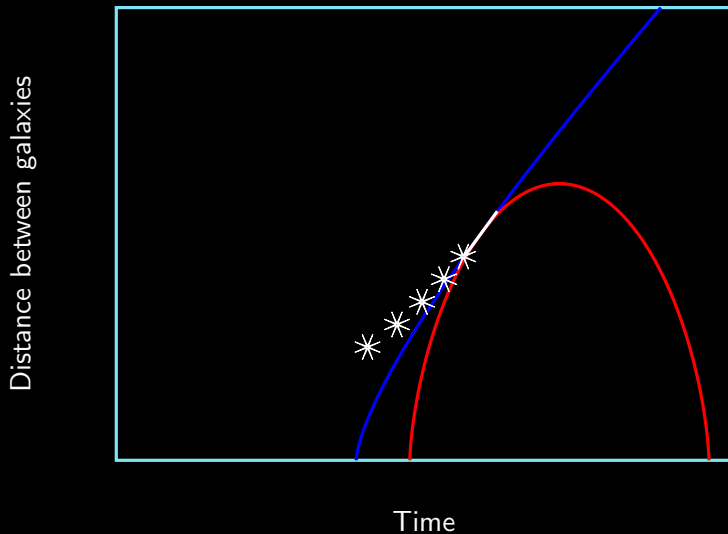
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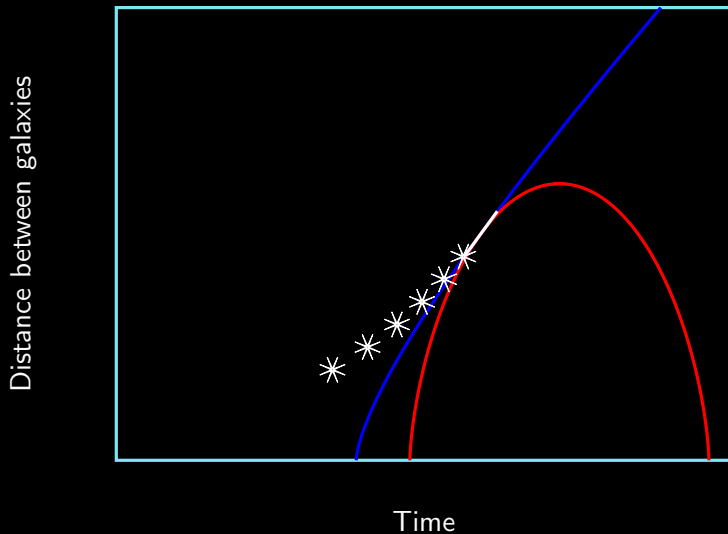
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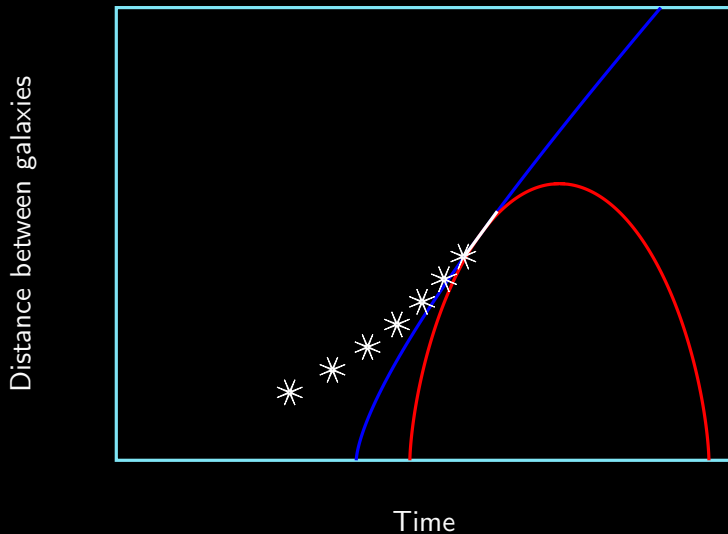
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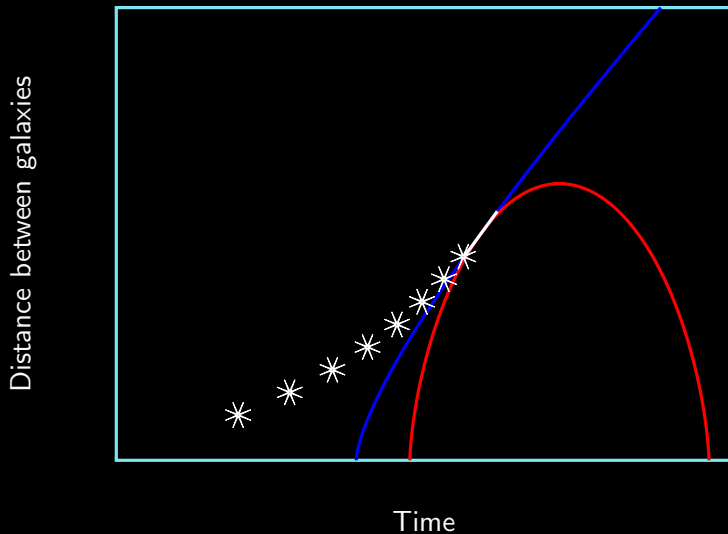
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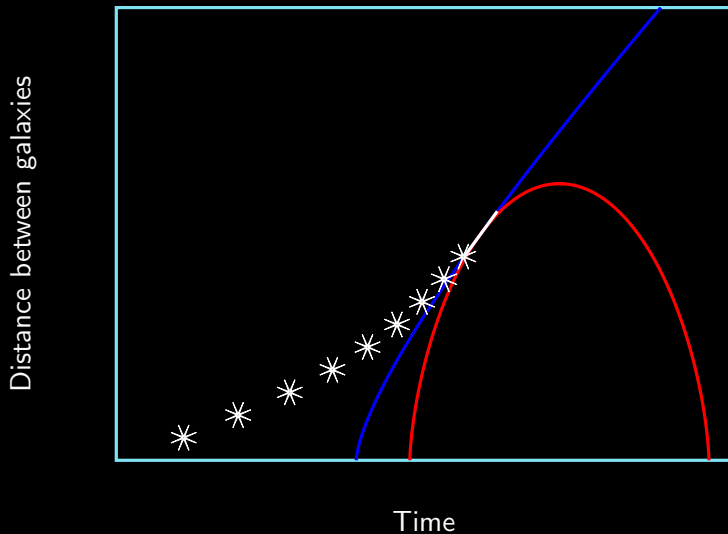
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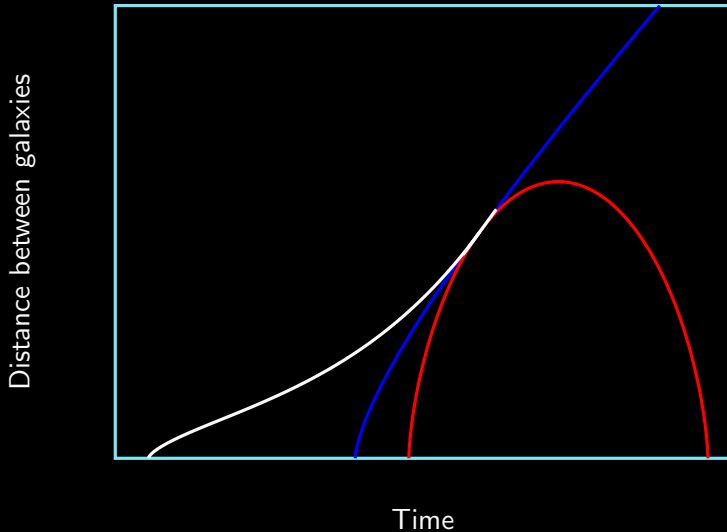
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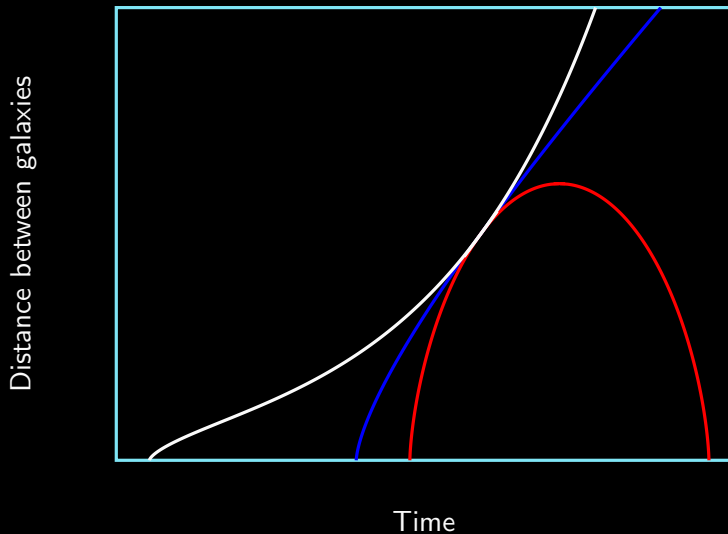
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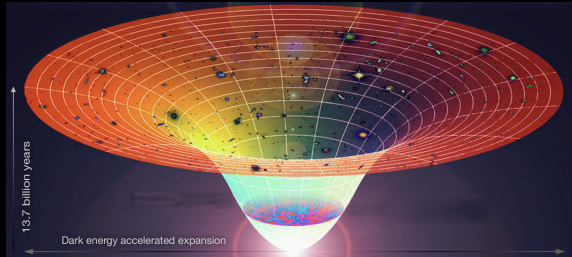
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Accelerated expansion

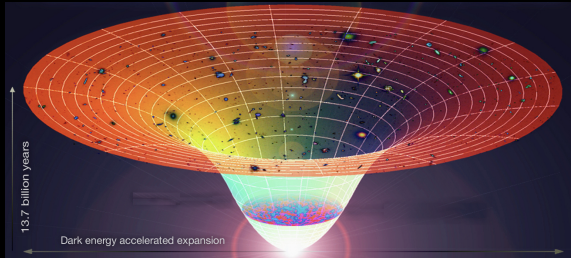
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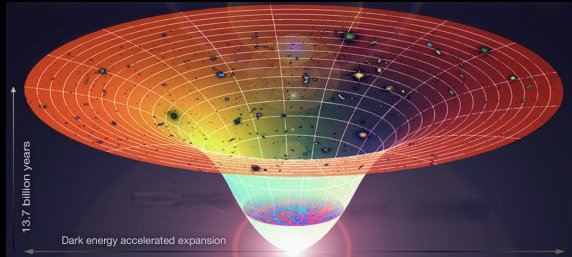
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Accelerated expansion

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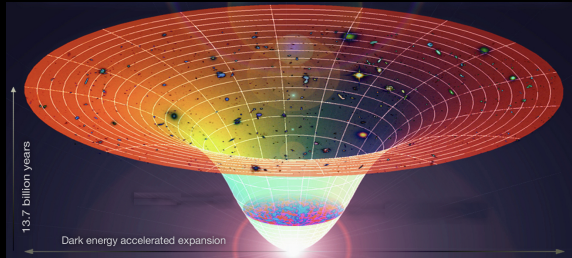
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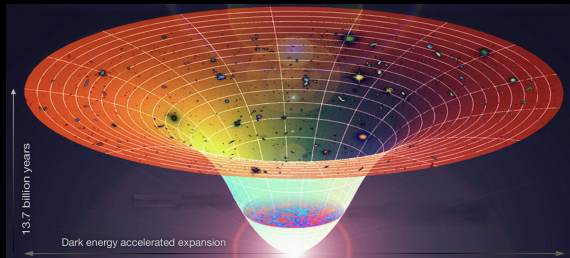
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- ▶ At some point in the past (≈ 4 bya) the universe's expansion started speeding up!
- ▶ In stark contrast to what we predicted.
- ▶ As if some “force” is driving the increased expansion.
- ▶ Or, as if space itself has some inherent “springyness” pushing everything apart.



Dark Energy

- Dark energy is a form of energy that is hypothesized to be responsible for the observed acceleration of the universe's expansion.
- It is called "dark" because it does not interact with light or other forms of electromagnetic radiation, making it invisible to telescopes.
- Dark energy is thought to make up about 68% of the total energy density of the universe.
- The existence of dark energy was first inferred from observations of distant supernovae in the late 1990s, which showed that the universe's expansion is accelerating.
- The leading theoretical explanation for dark energy is the cosmological constant, a term introduced by Albert Einstein in his equations of general relativity.
- Other theories propose that dark energy could be a new form of energy or a modification of gravity.
- Despite its name, dark energy is not necessarily "dark" in the sense of being black or absorbing light; it simply does not interact with light in a way that allows it to be directly observed.
- The study of dark energy is a major focus of modern cosmology, and understanding its nature could revolutionize our understanding of the universe's structure and evolution.

Dark Energy

- ▶ We attribute this accelerated expansion to a quantity we call *Dark Energy*.

What actually is Dark Energy?

- ▶ Short answer:

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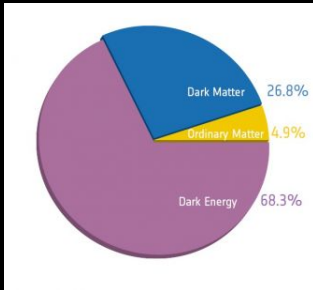
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Anything producing a large scale, diffuse, (negative) pressure.

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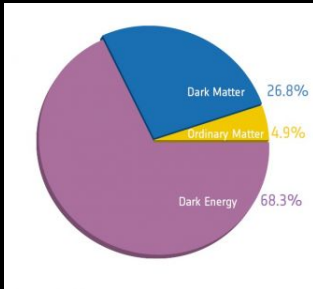
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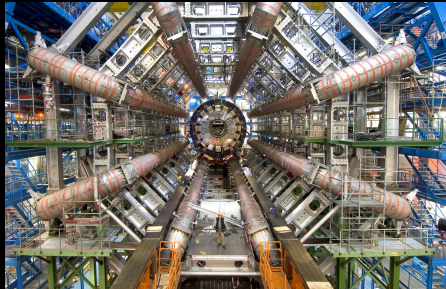
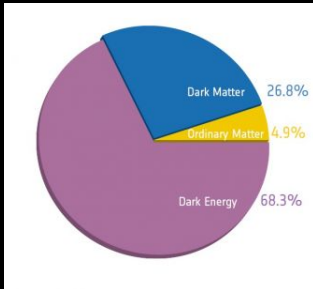
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What could this all pervading energy be?

Common interpretation

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- ▶ Higgs Field/Higgs Boson?

What could this all pervading energy be?

Common interpretation

- ▶ Dark energy is likely to be some form of quantum field, generating a universal expansion force.
- ▶ Higgs Field/Higgs Boson?
- ▶ Nope. Factor of 10^{120} too much energy in Higgs field.

What could this all pervading energy be?

Common interpretation

- [illegible]

Dark Energy...

...or Cosmological Constant?

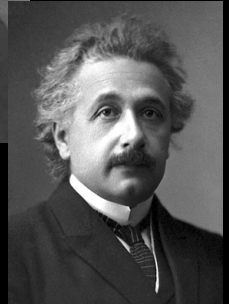
- ▶ Einstein's gravity can be repulsive at large scales.



Dark Energy...

...or Cosmological Constant?

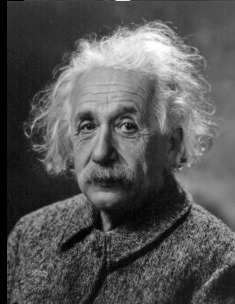
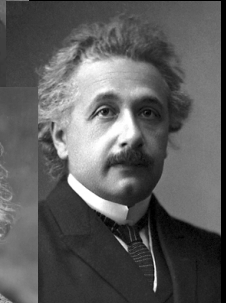
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Dark Energy...

...or Cosmological Constant?

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Dark matter and Dark Energy as a prediction

In general, if you encounter a discrepancy between theory and observation, there are two options:

Dark matter and Dark Energy as a prediction

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Dark matter and Dark Energy as a prediction

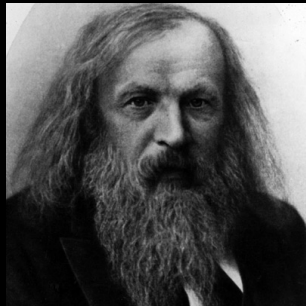
In general, if you encounter a discrepancy between theory and observation, there are two options:

- ▶ Modify the theory
- ▶ Predict a new object

Example 1.

The Periodic Table

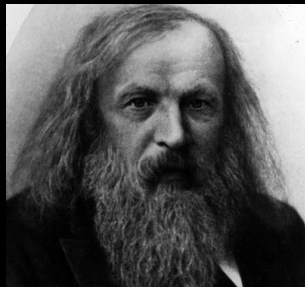
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.



Example 1.

The Periodic Table

- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.

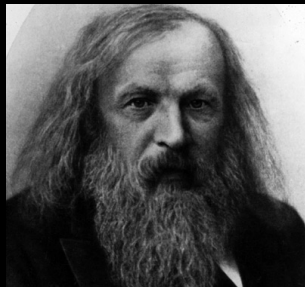


	Gruppe I. — H ⁰	Gruppe II. — H ⁰	Gruppe III. — H ⁰	Gruppe IV. H ¹ H ⁰	Gruppe V. H ¹ H ⁰	Gruppe VI. H ¹ H ⁰	Gruppe VII. H ¹ H ⁰	Gruppe VIII. — H ⁰
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,5	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Ca=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Str=87	Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ra=104, Rn=104, Po=106, Ag=108.
7	(Ag=108)	Cd=112	In=115	Su=118	Sh=122	Tu=125	J=127	
8	Cs=133	Ba=137	Pd=138	Ce=140	—	—	—	
9	(—)	—	—	—	—	—	—	
10	—	—	—	—	—	—	—	
11	(Au=197)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	

Example 1.

The Periodic Table

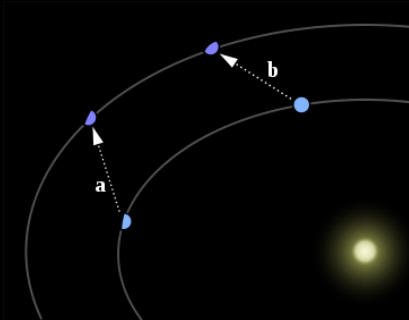
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.
- ▶ When these elements were discovered, he was hailed as a scientific hero.



Atom. N°	Gruppe I. — H ⁰	Gruppe II. — H ⁰	Gruppe III. — R ⁰	Gruppe IV. — R ⁰	Gruppe V. — R ⁰	Gruppe VI. — R ⁰	Gruppe VII. — R ⁰	Gruppe VIII. — R ⁰
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8	Ce=123	Ba=127	YDi=128	Po=140	—	—	—	
9	(—)	—	—	—	—	—	—	
10	—	—	YEr=178	La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Ti=204	Pb=207	Bism=208	—	—	
12	—	—	—	Th=231	—	U=240	—	

Example 2.

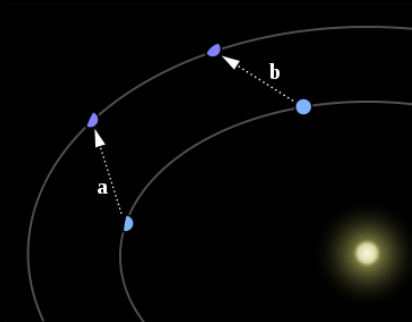
Orbits of Uranus and Mercury



- In 1841, it was noticed that Uranus' orbit was slightly off from predictions.

Example 2.

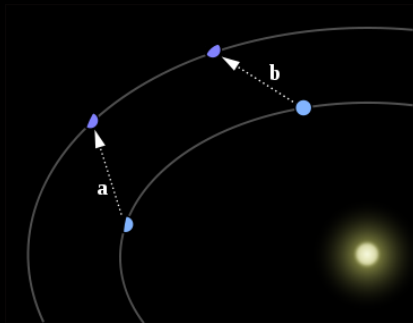
Orbits of Uranus and Mercury



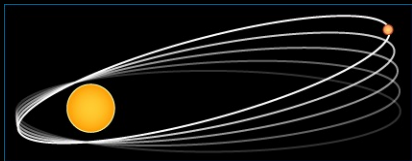
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- ▶ This observation led to the discovery of Neptune in 1846.

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Orbits of Uranus and Mercury



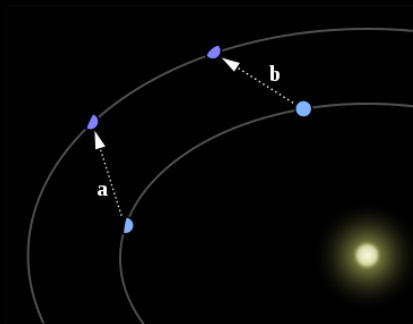
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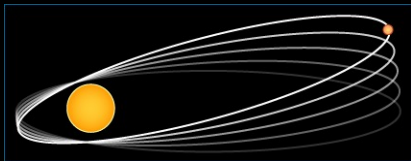
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Example 2.

Orbits of Uranus and Mercury



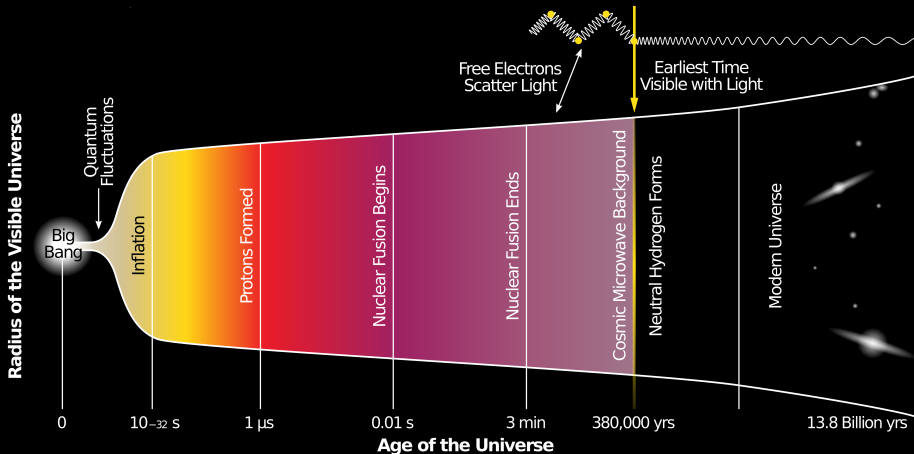
- ▶ In 1841, it was noticed that Uranus' orbit was slightly off from predictions.
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- ▶ In 1859, a similar problem emerged with the orbit of mercury
- ▶ This issue wasn't solved until Einstein's theory of general relativity superseded Newton's gravity in 1915.

The beginning of time

$t = 0$



The dawn of a new era in astronomy

The rise of the mega-telescope

The dawn of a new era in astronomy

The rise of the mega-telescope

- ▶ Euclid Satellite

The dawn of a new era in astronomy

The rise of the mega-telescope

- ▶ Euclid Satellite
- ▶ European Extremely Large Telescope

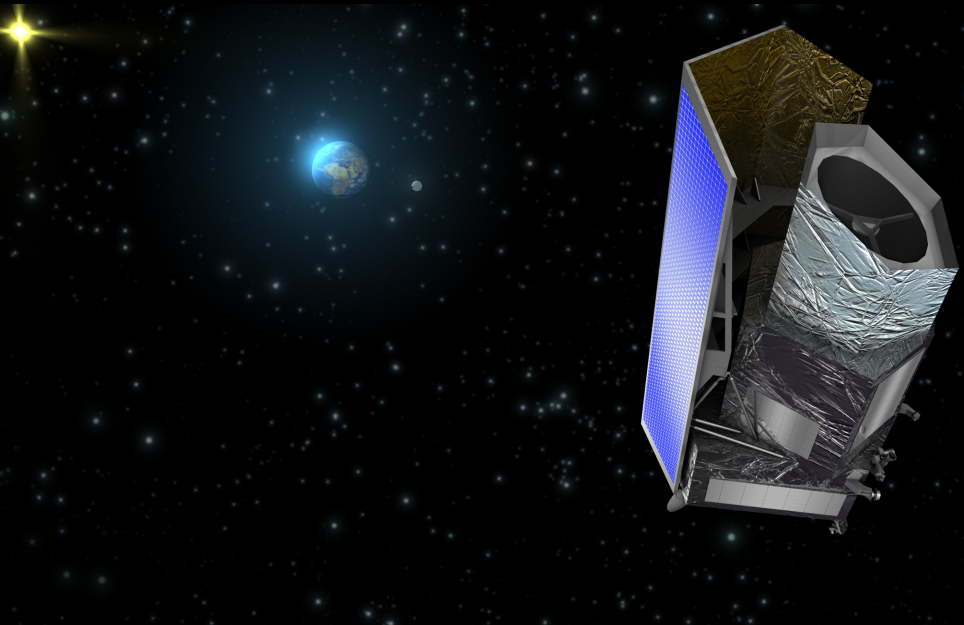
The dawn of a new era in astronomy

The rise of the mega-telescope

- ▶ Euclid Satellite
- ▶ European Extremely Large Telescope
- ▶ Square Kilometre Array

The Euclid Satellite

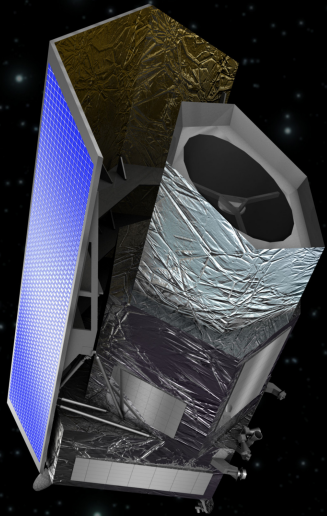
Exploring dark energy



The Euclid Satellite

Exploring dark energy

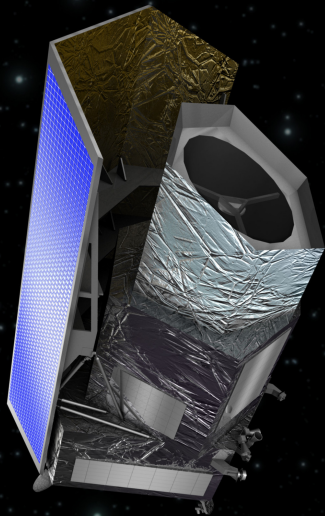
► Launch date $\approx$ 2020



The Euclid Satellite

Exploring dark energy

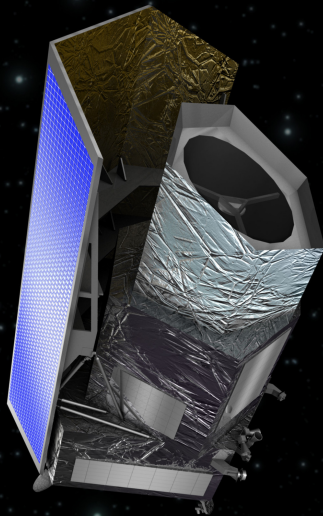
- ▶ Launch date $\approx$ 2020
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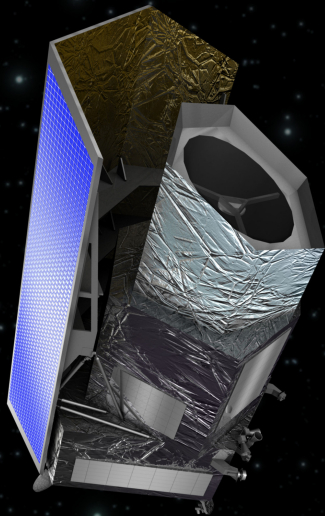
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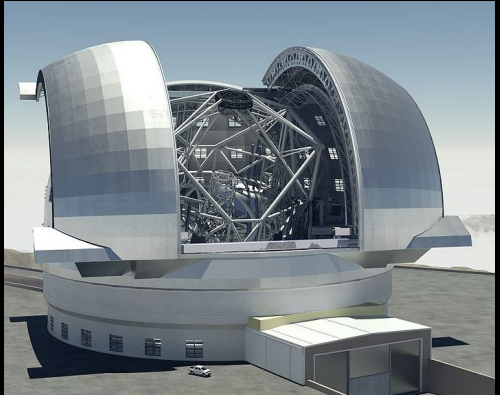
Exploring dark energy

- ▶ Launch date $\approx$ 2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.



European Extremely Large Telescope (E-ELT)

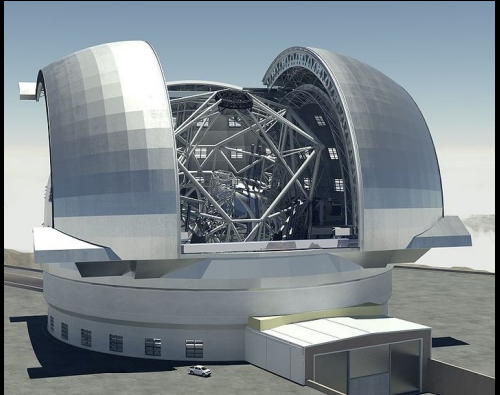
2024: Ground-based 40m optical telescope



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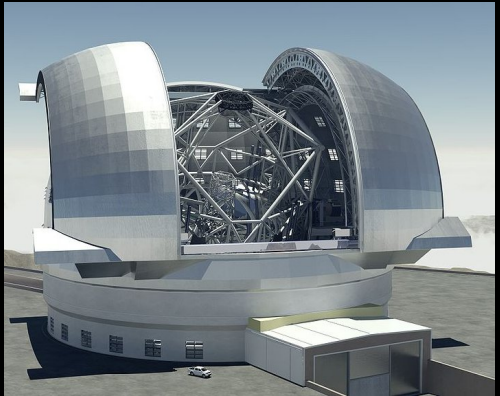
► Cost $\approx$ €1.3bn



European Extremely Large Telescope (E-ELT)

2024: Ground-based 40m optical telescope

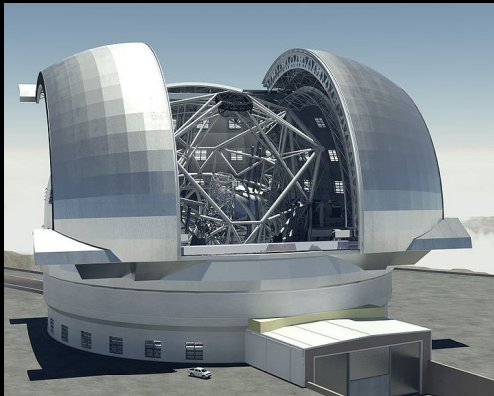
- ▶ Cost $\approx$ €1.3bn
- ▶ Search for extrasolar planets.



European Extremely Large Telescope (E-ELT)

2024: Ground-based 40m optical telescope

- ▶ Cost $\approx$ €1.3bn
- ▶ Search for extrasolar planets.
- ▶ Over a 20 year baseline, it will be sufficiently accurate to watch the universe expand.



Square Kilometre Array (SKA)

2019: World-wide radio telescope



SPDO / Square Kilometre Array Project

Square Kilometre Array (SKA)

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► Cost: €1.5bn

Square Kilometre Array (SKA)

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- ▶ Cost: €1.5bn
- ▶ Testing GR, DE, DM, Magnetism...

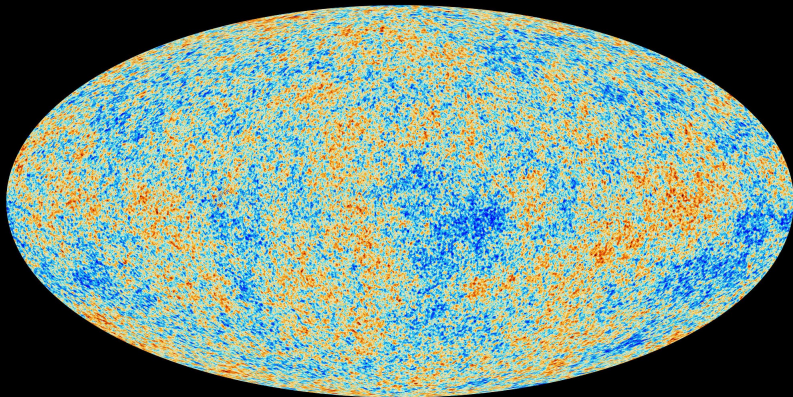
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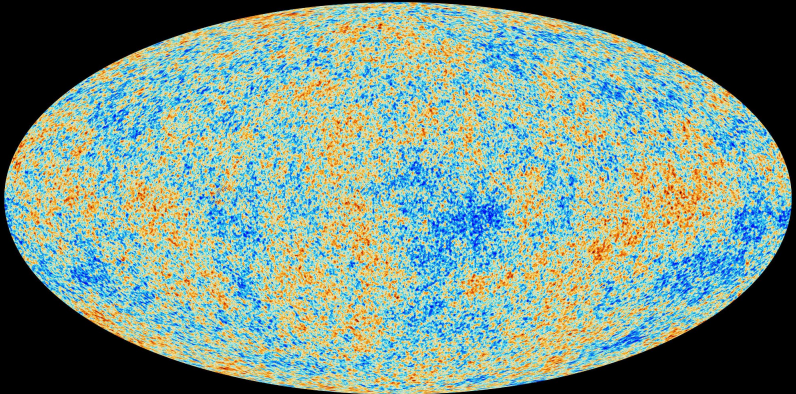


- ▶ Cost: €1.5bn
- ▶ Testing GR, DE, DM, Magnetism...
- ▶ Imaging Hydrogen through the universe will build a 3D picture of the first ripples of structure.

Summary

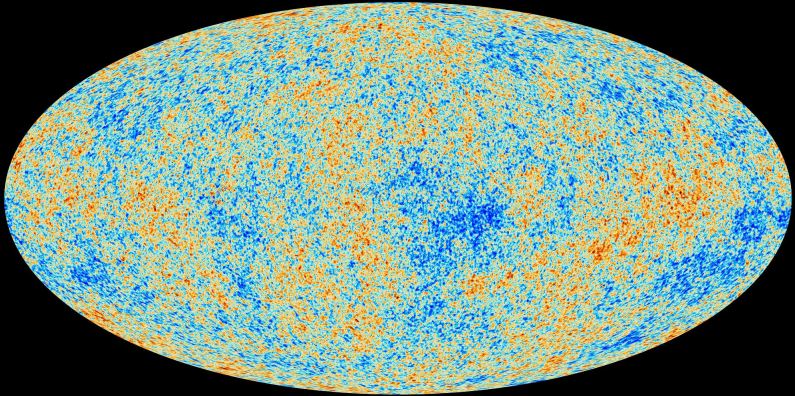


Summary



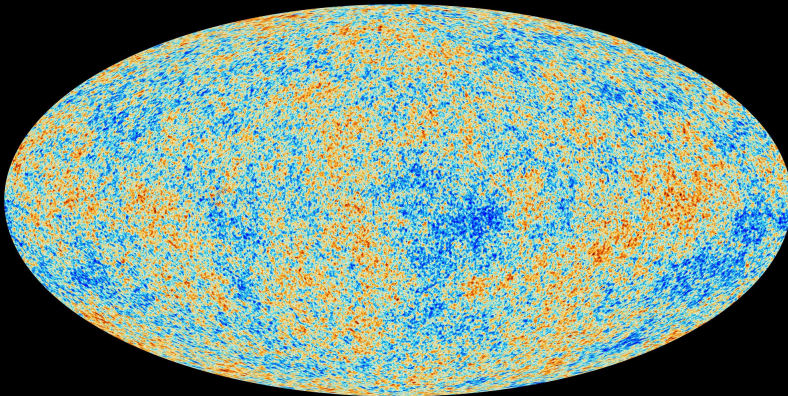
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Summary



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Summary



- ▶ Modern Astrophysics sits in a very privileged position:
- ▶ There are many unsolved theoretical problems,
- ▶ And a wealth of data to explore them with.

What is Cosmology?

Cosmology is the study of the universe as a whole

Age: How old is the universe?

Birth: Is there a beginning of time? What was it like?

Fate: Will everything come to an end?

Size: How big is the cosmos? Is there an edge?

Shape: What happens as you travel across the universe?

Content: What are Dark Matter and Dark Energy?